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# Exact Selective Locality Reduction for Resource-Constrained Quantum Approximate Optimization

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**ABSTRACT** Abstract to be drafted. The abstract must be a single paragraph of 150 to 250 words, self-contained, without abbreviations, footnotes, references, displayed equations, or tables, and should include three or four keywords or phrases.

**INDEX TERMS** Pseudo-Boolean optimization, quadratization, quantum approximate optimization algorithm, quantum circuit compilation, resource-aware quantum algorithms, warm-start QAOA.

**TABLE 1.** Comparison of the proposed framework with the closest prior representation strategies (to be completed).

| Strategy | Exact | Aux. | Locality | Connectivity | Selection |
|---|---|---|---|---|---|
| Native higher-order | ✓ | 0 | $\deg f$ | – | – |
| Full exact quadratization | ✓ | max | 2 | – | count |
| Hardware-aware aux. selection | ✓ | max | 2 | ✓ | degree |
| Approximate quadratization | – | var. | 2 | – | fidelity |
| This work | ✓ | var. | mixed | ✓ | multi-obj. |

## I. INTRODUCTION

HIGHER-ORDER binary interactions arise naturally in clause penalties, hypergraph objectives, and many-body spin models. Text to follow.

## II. RELATED WORK

### A. QUADRATIZATION OF PSEUDO-BOOLEAN FUNCTIONS

### B. HIGHER-ORDER ENCODINGS AND QAOA CIRCUIT SYNTHESIS

### C. HARDWARE-AWARE REFORMULATION AND COMPILATION

### D. WARM STARTS AND INITIALIZATION

## III. EXACT SELECTIVE LOCALITY REDUCTION

### A. SETTING AND NOTATION

### B. SHARP AGGREGATE-RESIDUAL PENALTY

*Proposition 1 (Sharp aggregate-residual penalty):* Statement to follow.

### C. SEQUENTIAL COMPOSITIONAL EXACTNESS

*Theorem 1 (Sequential compositional exactness):* Statement to follow.

### D. CUBIC CLOSED FORM AND PAIR REUSE

### E. WORKED MAX-3SAT EXAMPLE

## IV. RESOURCE-AWARE REPRESENTATION SELECTION

### A. COMPILATION AND CIRCUIT MODEL

### B. RESOURCE SCORE, BUDGETS, AND FIBRE DIAGNOSTIC

### C. PARETO BEAM SEARCH

### D. BRANCH-AND-BOUND CERTIFICATION

### E. REPRESENTATION-CONSISTENT WARM STARTS

## V. EXPERIMENTAL EVALUATION

### A. BENCHMARK FAMILIES AND INSTANCE TIERS

### B. REPRESENTATIONS, BASELINES, AND COMPILATION PROTOCOL

### C. QAOA PROTOCOL AND STATISTICAL TREATMENT

### D. EXACTNESS AND SELECTOR VALIDATION

### *E. COMPILED RESOURCES ACROSS COUPLING MAPS*

### *F. FINITE-DEPTH QAOA UNDER MATCHED BUDGETS*

### *G. WARM-START PERFORMANCE*

## VI. DISCUSSION

## VII. CONCLUSION

## ACKNOWLEDGMENT

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## DATA AND CODE AVAILABILITY

Processed data, source manifests, and the code used to generate and analyze the reported results are available in the project repository. The complete raw result archive will be deposited with a persistent identifier before publication.

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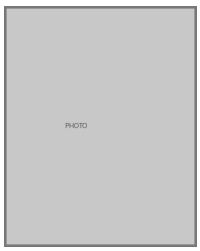
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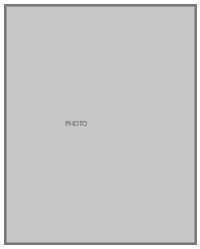
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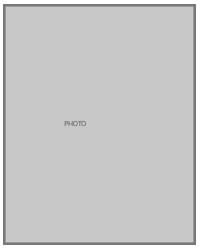
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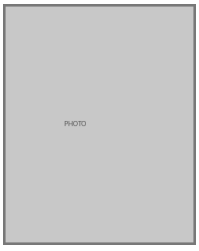

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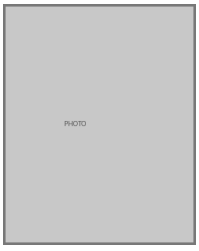

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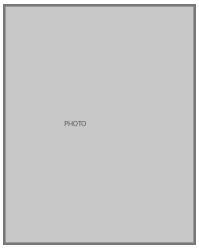
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